
Mixing Matters: Evidence Position Bias Across Attention and State-Space Language Models

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Abstract

1 Language models often answer multi-document questions less accurately when
2 the relevant document is placed in the middle rather than near the beginning or
3 end. We test whether this pattern differs between models that use attention and
4 models from the Mamba family, which process context through a running state.
5 For each question, we move the answer-bearing document through ten positions
6 while keeping the question and nine distracting documents fixed. Our main com-
7 parisons use 800 exploratory questions; smaller task and prompt checks use 50 or
8 200 examples, and the larger held-out set has not been examined. Pythia-2.8B was
9 about five percentage points more accurate when the answer appeared near the be-
10 ginning than in the middle, whereas two similarly sized, Pile-trained Mamba mod-
11 els showed no such beginning advantage. A more closely matched 8B comparison
12 pointed in the same direction: a hybrid Mamba-2 model with a small number of at-
13 tention layers showed a larger beginning advantage than the pure Mamba-2 model,
14 but the difference between them remains too uncertain for a firm conclusion. The
15 contrast appears in larger but not the smallest model pairs, is not explained by the
16 tested change of training corpus, and reappears for Pythia on a synthetic retrieval
17 task. Prompt changes can strongly alter the pattern, and exploratory internal-state
18 analyses show that Mamba retains position information even without the begin-
19 ning advantage. These model-family differences are not causal evidence about
20 sequence mixing: at the 2.8B scale, depth, RoPE usage, and recurrent-state de-
21 sign and size remain confounded.

22 1 Introduction

23 Transformers often read multi-document contexts unevenly: accuracy peaks when the answer-
24 bearing document is near an edge and falls in the middle (Liu et al., 2024). This behavior matters
25 because retrieval systems cannot assume that all evidence positions are equally usable. Long-context
26 benchmarks expose related failures across question answering, summarization, and retrieval (Bai et
27 al., 2024; Shaham et al., 2023; Li et al., 2024). Prior work links lost-in-the-middle behavior to
28 U-shaped attention patterns (Hsieh et al., 2024a), but our near-2.8B model-family comparison still
29 confounds sequence mixing with depth, RoPE usage, recurrent-state design and size, and checkpoint
30 family (Su et al., 2024; Press et al., 2022). State-space models such as Mamba (Gu and Dao, 2024;
31 Dao and Gu, 2024) provide a useful contrast because they compress the prefix into a recurrent state
32 instead of using dense causal attention.

33 We make three contributions. First, we build an auditable ten-position evaluation with floor and
34 ceiling anchors, preserved generations, and inference settings recorded per example. Second, we
35 find a paired difference in primacy profiles among Pile-trained models near 2.8B, subject to those
36 confounds, and examine its direction with a more tightly matched 8B pure/hybrid pair. Third, we

organize scale, corpus, task, prompt, sink, and probe results into mechanistic evidence with explicit uncertainty. All reported results use an exploratory split; the held-out confirmatory split remains unspent.

2 Related work

Position bias in long contexts Liu et al. (2024) established the lost-in-the-middle curve in retrieval and multi-document QA. This task derives its questions from Natural Questions (Kwiatkowski et al., 2019), while LongBench, ZeroSCROLLS, and LooGLE broaden long-context evaluation across tasks, languages, and reasoning demands (Bai et al., 2024; Shaham et al., 2023; Li et al., 2024). Subsequent work measures effective context length (Hsieh et al., 2024b), associates the curve with attention allocation (Hsieh et al., 2024a), and proposes reordering, compression, retrieval, and context-use interventions (An et al., 2024; Peysakhovich and Lerer, 2023; Jiang et al., 2024; Xu et al., 2024; Agrawal et al., 2024). Irrelevant context can itself distract a model, motivating our use of the same distractors at every tested position (Shi et al., 2023). We focus on whether the curve changes across sequence mixers.

Sequence mixers Dense self-attention (Vaswani et al., 2017) mixes token pairs under a causal mask and commonly encodes order with rotary embeddings or linear attention biases (Su et al., 2024; Press et al., 2022). Alternative long-sequence mixers include structured state spaces, recurrent weighted key-value models, and long convolutions (Gu et al., 2022; Peng et al., 2023; Poli et al., 2023). Mamba and Mamba-2 carry information through selective state-space recurrences (Gu and Dao, 2024; Dao and Gu, 2024). Hybrid models such as Griffin, Jamba, and Nemotron-H combine recurrent or state-space blocks with attention (De et al., 2024; Lieber et al., 2024; NVIDIA, 2025). Waleffe et al. (2024) release the pure and hybrid 8B Mamba-2 checkpoints used in our tightest controlled comparison.

Attention sinks and probes Xiao et al. (2024) showed that causal transformers can concentrate attention on initial tokens, and Sun et al. (2024) connected unusually large internal activations to a small set of tokens. We test whether this attention-sink statistic covaries with primacy, treating the result as correlational because a valid sink-blocking ablation is absent. Our position probe follows the principle that a diagnostic classifier needs a control task before its accuracy can be interpreted as evidence about a representation (Hewitt and Liang, 2019).

3 Measurement and statistics

Task and anchors We use the 2,655 released questions from the Lost-in-the-Middle 10-document QA setting (Liu et al., 2024), which is built from Natural Questions (Kwiatkowski et al., 2019). Each question has one gold document and nine distractors. We build ten prompts per question, moving only the gold document through positions 0 to 9 while retaining the same distractors and token budget. A no-gold floor measures guessing or memorized knowledge, and a gold-only ceiling measures answerability under perfect retrieval.

Edges and statistics Let a_S , a_M , and a_E be mean accuracy at start positions $\{0, 1\}$, middle positions $\{4, 5\}$, and end positions $\{8, 9\}$. We define primacy as $a_S - a_M$ and recency as $a_E - a_M$. The question is the unit of analysis. Each of 10,000 bootstrap samples draws complete ten-position question bundles with replacement, following the bootstrap principle of resampling the independent unit (Efron, 1979), and model comparisons reuse the same sampled questions. We report percentile 95% confidence intervals and apply Holm’s sequential correction to the primacy and recency tests within each comparison family (Holm, 1979). The fixed split contains 800 exploratory and 1,855 confirmatory questions. Only the exploratory split is reported.

Reproducibility and controls The harness vendors the official prompt builder and best_subspan_em scorer at a pinned commit. It also records normalized exact match, following the normalization convention popularized by SQuAD (Rajpurkar et al., 2016), and first-line extraction. Generation is greedy with temperature 0, at most 32 new tokens, fixed seeds, BF16 where supported, pinned model revisions, and one Mamba execution path. Each unique run preserves prompts, raw

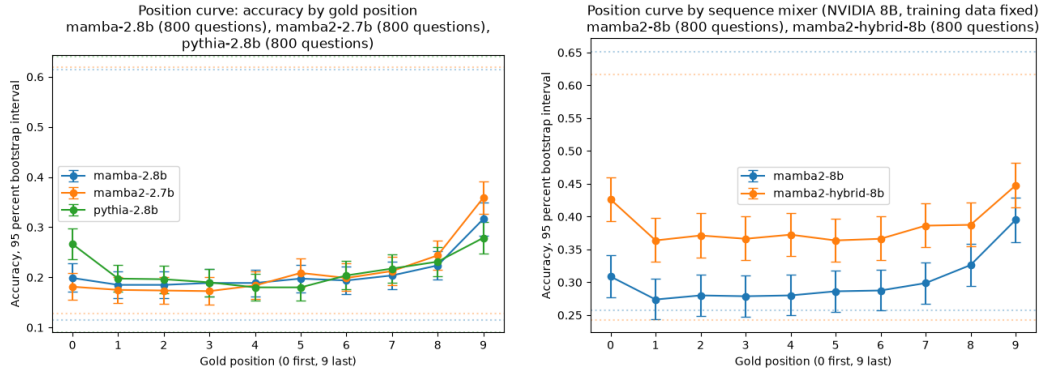
generations, failures, token counts, software versions, and decoding settings. The released key-value task provides an executed positive control: the pipeline detects a +0.38 Pythia start-versus-middle effect. On 200 exploratory Pythia questions, sham-gold accuracy was 0.1235, within 0.0285 of the 0.0950 closed-book floor; primacy was +0.0050 (95% CI $[-0.0325, 0.0425]$) and recency was -0.0175 (95% CI $[-0.0500, 0.0125]$). The distractor-order control’s permutation accuracies were 0.2850, 0.3000, and 0.2767, a maximum spread of 0.0233. Both controls passed their prespecified gates on Pythia; they do not test the 8B Megatron checkpoints.

4 Architecture comparisons

Primary 2.8B comparison We compare pythia-2.8b (Biderman et al., 2023) with mamba-2.8b and mamba2-2.7b. All are base models pretrained on the Pile (Gao et al., 2020), have similar parameter counts, and use related tokenizers. Table 1 and Figure 1 show a positive primacy edge for Pythia and no positive primacy edge for either Mamba model. The paired Mamba-minus-Pythia primacy interactions are -0.053 and -0.070 (both Holm $p < 10^{-4}$), while Mamba-1 versus Mamba-2 is uncertain ($+0.017$, 95% CI $[-0.005, 0.038]$). All three models have positive recency edges. These comparisons match corpus and approximate scale, but not depth, positional encoding, state size, or checkpoint family.

Table 1: Pile-trained models on 800 exploratory questions. Intervals use a question-level paired bootstrap; p is Holm-corrected.

Model	Primacy (95% CI)	Recency (95% CI)	Floor / Ceiling
pythia-2.8b	$+0.052$ $[0.032, 0.072]$, $p < 10^{-4}$	$+0.075$ $[0.054, 0.097]$, $p < 10^{-4}$	0.091 / 0.640
mamba-2.8b	-0.001 $[-0.016, 0.014]$, $p = 0.91$	$+0.077$ $[0.058, 0.097]$, $p < 10^{-4}$	0.115 / 0.615
mamba2-2.7b	-0.018 $[-0.034, -0.003]$, $p = 0.02$	$+0.105$ $[0.086, 0.124]$, $p < 10^{-4}$	0.128 / 0.619



(a) Pile-trained models near 2.8B parameters

(b) Matched pure and hybrid Mamba-2 models at 8B

Figure 1: Accuracy as the answer-bearing document moves through the context on 800 exploratory questions. Pythia rises at the beginning, while both smaller Mamba variants do not; the matched 8B hybrid also rises more than the pure model, but their paired difference remains uncertain. Dotted lines show each model’s no-gold floor and gold-only ceiling.

Matched 8B attention comparison Our strongest control uses NVIDIA’s pure and hybrid 8B Mamba-2 checkpoints (Waleffe et al., 2024), which share training data, tokenizer, scale, depth, and positional setup. The hybrid contains 7% attention layers. Within models, the pure model’s primacy edge is uncertain ($+0.008$, $p = 0.36$) and the hybrid’s is positive ($+0.027$, $p = 0.007$). That difference of significance is not itself a significant difference. The paired hybrid-minus-pure primacy effect is $+0.019$ (95% CI $[-0.006, 0.044]$, Holm $p = 0.144$), while the recency effect is -0.028 (95% CI $[-0.055, -0.001]$, Holm $p = 0.089$). This direction supports the 2.8B observation but does not isolate a causal attention effect.

Table 2: Matched 8B primacy comparison on 800 exploratory questions. The paired interaction is the relevant architecture test.

Contrast	Primacy	95% CI	Holm p
pure Mamba-2	+0.008	$[-0.009, 0.024]$	0.362
hybrid	+0.027	$[0.008, 0.047]$	0.007
hybrid minus pure	+0.019	$[-0.006, 0.044]$	0.144

5 Scope checks

Scale Across five approximate size pairs, the Pythia-minus-Mamba primacy gap is near zero at the two smallest scales and is +0.062, +0.069, and +0.053 in the three larger pairs (Figure 2). The smallest Pythia ceiling is only 0.009, so this is a descriptive capability-and-family trend, not evidence for a depth threshold.

Training corpus Holding the Mamba-2.8B architecture fixed, the Pile-minus-SlimPajama primacy effect is -0.007 (95% CI $[-0.030, 0.016]$, $p = 0.568$) and the recency effect is -0.016 (95% CI $[-0.044, 0.012]$, Holm $p = 0.502$) (Soboleva et al., 2023). The floor differs substantially (0.270 versus 0.114), so this comparison finds no edge-shape change but does not rule out training-data effects (Figure 2).

Synthetic retrieval On 50 RULER niah_single_1 examples at 2K tokens (Hsieh et al., 2024b), Pythia again has a primacy edge (+0.12, 95% CI $[0.05, 0.20]$). Both Mamba models score 1.0 at every depth, so this check reproduces the transformer pattern but cannot compare mixer curves on the synthetic task (Figure 3).

Descriptive system comparison We also run Llama-3.1-8B (Dubey et al., 2024), Qwen2.5-7B (Yang et al., 2024), and Nemotron-H-8B (NVIDIA, 2025). All three show positive primacy edges (+0.076, +0.041, +0.067). Only Llama versus Qwen has a significant paired primacy interaction (+0.036, Holm $p = 0.016$). Because the systems differ on architecture, data, tokenizer, token count, depth, and positional encoding, these results establish prevalence, not an architectural explanation (Figure 4).

6 Mechanistic evidence and intervention

Attention sinks track primacy Across the Pythia family, the 160M model has almost no token-0 attention sink (final-layer share 0.003), while the 2.8B model reaches 0.455 (Xiao et al., 2024). Pythia-410M has a small final-layer sink (0.034) and no significant primacy edge. Figure 3 is consistent with a sink mechanism, but the measurements share model scale as a confound and no valid sink ablation was completed.

Utilization, not storage, is a hypothesis A balanced linear probe on frozen mid-depth states predicts edge versus middle position above its shuffled-label control for Pythia (0.603 versus 0.484) and Mamba (0.648 versus 0.508). Thus position information remains linearly decodable even when Mamba has no positive primacy edge. This supports a utilization hypothesis, but probe decodability does not prove that the model stores or uses the answer-bearing content.

Training-free prompt intervention On 200 exploratory questions, repeating the question both before and after the documents increases Mamba primacy from 0.000 to +0.103 ($p < 10^{-4}$). Placing the question only before the documents does not create primacy (+0.020, $p = 0.366$), but reduces recency from +0.070 to +0.010. The intervention is training-free and practically useful, but it changes query repetition and token placement together, so it is not a clean test of fixed-state compression.

Prompt sensitivity Changing the prompt template reduces Pythia primacy from +0.070 to +0.005 in the 200-question mechanism subset. This result warns against treating the measured magnitude as a template-invariant architectural constant.

7 Discussion and limitations

The most secure result is the 2.8B exploratory architecture-position interaction. The matched 8B direction, sink correlation, and probe results jointly make attention-mediated utilization a plausible hypothesis, while prompt sensitivity shows that architecture is not the only determinant.

Several limitations set the confirmatory boundary. All analyses use the exploratory split, so the planned 1,855-question test must be run under a frozen analysis before confirmatory language is warranted. The 2.8B and scale comparisons retain depth, positional-encoding, state-size, and capability confounds. The matched 8B paired contrast crosses zero after Holm correction, and its run manifest records a dirty source tree, weakening exact reproducibility. These certification controls passed on Pythia, but they do not test the 8B Megatron checkpoints. A 50-generation blinded audit sample exists, but human labels were not completed. Finally, the attempted hybrid sink block did not alter the model because its custom attention path did not honor the hook; it is not an ablation result. The next decisive study should freeze exclusions and contrasts, repair the hybrid intervention, and evaluate the held-out split once.

8 Conclusion

On 800 exploratory Lost-in-the-Middle questions, Pythia-2.8B and two Pile-trained Mamba models have different primacy profiles under a paired ten-position evaluation. A matched 8B comparison is directionally consistent but statistically uncertain, so the present evidence supports an architecture-position interaction without establishing that attention causes it. The held-out split and the remaining control work provide a clear path from this exploratory workshop result to a confirmatory study.

A Experiment artifacts

The repository preserves raw generations, environment records, machine-readable summaries, and plots for every reported phase. The panels below reproduce the exact committed artifact images without redrawing or smoothing them. They summarize scope and mechanism checks that support interpretation but do not upgrade the exploratory results to confirmatory or causal evidence.

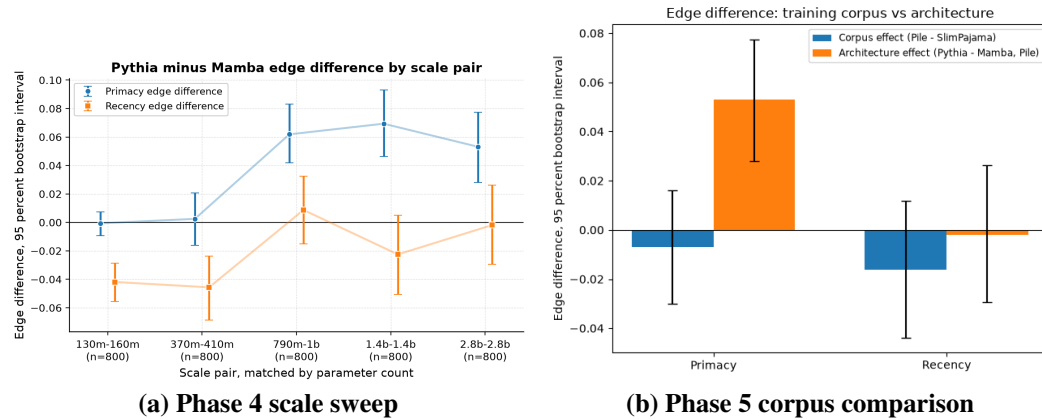


Figure 2: Exploratory scope checks. The scale sweep is descriptive because model capability and family differences vary together. The corpus comparison holds the Mamba architecture fixed and finds no clear change in curve shape, despite a substantial change in overall accuracy.

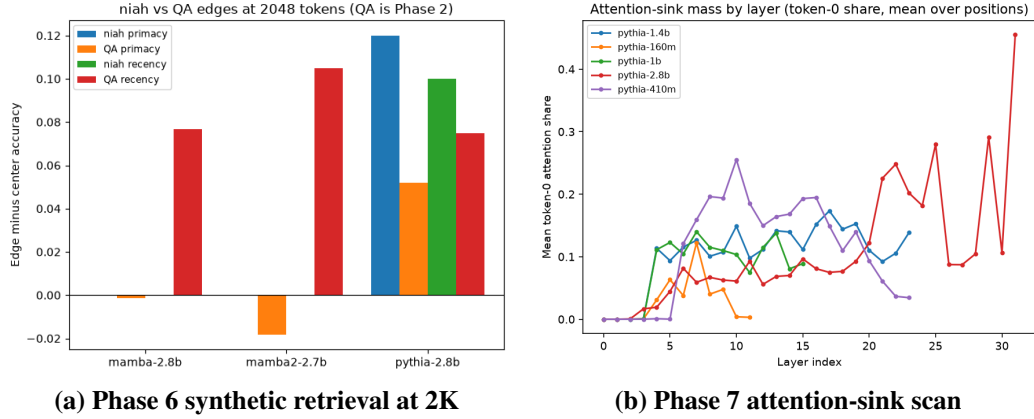


Figure 3: Exploratory task and mechanism artifacts. Pythia has a middle dip on the 2K synthetic task, while both Mamba models saturate and therefore cannot reveal a comparable curve. Late-layer token-0 attention grows with Pythia scale and covaries with the beginning advantage, but scale is a shared confound and no valid causal ablation was completed.

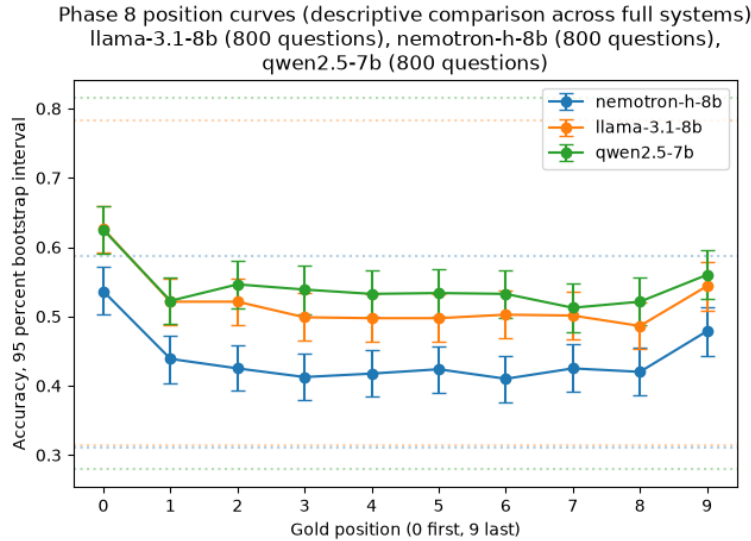


Figure 4: Phase 8 descriptive production-system comparison. All three systems show a beginning advantage, but differences in architecture, data, tokenizer, training, and capability prevent attribution to a single design choice.

References

- D. Agrawal, S. Gao, and M. Gajek. Can't remember details in long documents? You need some R&R. In *Findings of the Association for Computational Linguistics: EMNLP*, pages 12692-12704, 2024.
- C. An, S. Huang, J. Zhang, et al. Make your LLM fully utilize the context. In *Advances in Neural Information Processing Systems*, 2024.
- Y. Bai, X. Lv, J. Zhang, et al. LongBench: A bilingual, multitask benchmark for long context understanding. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, pages 3119-3137, 2024.
- S. Biderman, H. Schoelkopf, Q. Anthony, et al. Pythia: A suite for analyzing large language models across training and scaling. In *International Conference on Machine Learning*, pages 2397-2430, 2023.
- T. Dao and A. Gu. Transformers are SSMs: Generalized models and efficient algorithms through structured state space duality. In *International Conference on Machine Learning*, 2024.
- S. De, S. L. Smith, A. Fernando, et al. Griffin: Mixing gated linear recurrences with local attention for efficient language models. *arXiv:2402.19427*, 2024.
- A. Dubey, A. Jauhri, A. Pandey, et al. The Llama 3 herd of models. *arXiv:2407.21783*, 2024.
- L. Gao, S. Biderman, S. Black, et al. The Pile: An 800GB dataset of diverse text for language modeling. *arXiv:2101.00027*, 2020.
- B. Efron. Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7(1):1-26, 1979.
- A. Gu and T. Dao. Mamba: Linear-time sequence modeling with selective state spaces. In *Conference on Language Modeling*, 2024.
- A. Gu, K. Goel, and C. Ré. Efficiently modeling long sequences with structured state spaces. In *International Conference on Learning Representations*, 2022.
- J. Hewitt and P. Liang. Designing and interpreting probes with control tasks. In *Proceedings of EMNLP-IJCNLP*, pages 2733-2743, 2019.
- S. Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65-70, 1979.
- C.-P. Hsieh, S. Sun, S. Krizan, S. Acharya, D. Rekesh, F. Jia, and B. Ginsburg. Found in the middle: Calibrating positional attention bias improves long context utilization. In *Findings of the Association for Computational Linguistics: ACL*, pages 14982-14995, 2024.
- C.-P. Hsieh, S. Sun, S. Krizan, et al. RULER: What's the real context size of your long-context language models? In *Conference on Language Modeling*, 2024.
- H. Jiang, Q. Wu, X. Luo, et al. LongLLMLingua: Accelerating and enhancing LLMs in long context scenarios via prompt compression. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, pages 1658-1677, 2024.
- T. Kwiatkowski, J. Palomaki, O. Redfield, et al. Natural Questions: A benchmark for question answering research. *Transactions of the Association for Computational Linguistics*, 7:452-466, 2019.
- O. Lieber, B. Lenz, H. Bata, et al. Jamba: A hybrid Transformer-Mamba language model. *arXiv:2403.19887*, 2024.
- J. Li, M. Wang, Z. Zheng, and M. Zhang. LooGLE: Can long-context language models understand long contexts? In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, pages 16304-16333, 2024.
- N. F. Liu, K. Lin, J. Hewitt, A. Paranjape, M. Bevilacqua, F. Petroni, and P. Liang. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12:157-173, 2024.
- NVIDIA. Nemotron-H: A family of accurate and efficient hybrid Mamba-Transformer models. *arXiv:2504.03624*, 2025.
- A. Peysakhovich and A. Lerer. Attention sorting combats recency bias in long context language models. *arXiv:2310.01427*, 2023.

222 B. Peng, E. Alcaide, Q. Anthony, et al. RWKV: Reinventing RNNs for the Transformer era. In *Findings of the*
223 *Association for Computational Linguistics: EMNLP*, pages 14048-14077, 2023.

224 M. Poli, S. Massaroli, E. Nguyen, et al. Hyena hierarchy: Towards larger convolutional language models. In
225 *International Conference on Machine Learning*, volume 202 of PMLR, pages 28043-28078, 2023.

226 O. Press, N. A. Smith, and M. Lewis. Train short, test long: Attention with linear biases enables input length
227 extrapolation. In *International Conference on Learning Representations*, 2022.

228 P. Rajpurkar, J. Zhang, K. Lopyrev, and P. Liang. SQuAD: 100,000+ questions for machine comprehension of
229 text. In *Proceedings of EMNLP*, pages 2383-2392, 2016.

230 U. Shaham, M. Ivgi, A. Efrat, J. Berant, and O. Levy. ZeroSCROLLS: A zero-shot benchmark for long text
231 understanding. In *Findings of the Association for Computational Linguistics: EMNLP*, pages 7977-7989,
232 2023.

233 F. Shi, X. Chen, K. Misra, N. Scales, D. Dohan, E. H. Chi, N. Schärli, and D. Zhou. Large language models can
234 be easily distracted by irrelevant context. In *International Conference on Machine Learning*, volume 202 of
235 PMLR, pages 31210-31227, 2023.

236 D. Soboleva, F. Al-Khateeb, R. Myers, et al. SlimPajama: A 627B token cleaned and deduplicated version of
237 RedPajama. Cerebras technical report, 2023.

238 J. Su, Y. Lu, S. Pan, A. Murtadha, B. Wen, and Y. Liu. RoFormer: Enhanced transformer with rotary position
239 embedding. *Neurocomputing*, 568:127063, 2024.

240 M. Sun, X. Chen, J. Kolter, and Z. Liu. Massive activations in large language models. *arXiv:2402.17762*, 2024.

241 A. Vaswani, N. Shazeer, N. Parmar, et al. Attention is all you need. In *Advances in Neural Information*
242 *Processing Systems*, 2017.

243 R. Waleffe, W. Byeon, D. Riach, et al. An empirical study of Mamba-based language models.
244 *arXiv:2406.07887*, 2024.

245 G. Xiao, Y. Tian, B. Chen, S. Han, and M. Lewis. Efficient streaming language models with attention sinks. In
246 *International Conference on Learning Representations*, 2024.

247 A. Yang, B. Yang, B. Zhang, et al. Qwen2.5 technical report. *arXiv:2412.15115*, 2024.

248 F. Xu, W. Shi, and E. Choi. RECOMP: Improving retrieval-augmented LMs with compression and selective
249 augmentation. In *International Conference on Learning Representations*, 2024.

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